

Dr. Magdalena Aflori

Senior Researcher (Grade I) · PhD in Physics, Habilitation in Chemistry

Address Alea Gr. Ghica Vodă 41A, 700487 Iași, Romania

Phone +40 332 880 219

E-mail maflori@icmpp.ro

Nationality Romanian

RESEARCHER IDS

ORCID: 0000-0001-5919-221X

Scopus Author ID: 8397724900

Web of Science ResearcherID: C-4305-2011



Senior researcher with over 25 years in fundamental and applied research on polymer materials and surface engineering, and proven leadership of EU-funded projects. **Project Director of TalentPass (Horizon Europe) and POINGBIO (ERDF)**, with hands-on experience across the main EU funding instruments — **Horizon Europe, Framework Programmes (FP6), ERDF and ESF** — covering proposal development, consortium building, budgeting, implementation and reporting through the EU Funding & Tenders Portal. Strong scientific record (100+ Web of Science articles, h-index 24, ~1700 citations, 4 patents) and consistent experience in scientific peer review (Elsevier, Wiley, MDPI, CRC Press) and in regional Smart Specialisation (RIS3) evaluation bodies. **Project management certified**; available for roles in project coordination and evaluation for projects.

€18M+

EU funding led/co-managed

100+

Web of Science articles

23

h-index

4

patents (+2 pending)

EU & INTERNATIONAL FUNDING EXPERIENCE

As Project Director / Coordinator

TalentPass — Fostering European Talents for Widening Circular Economy

2025 – 2029

Horizon Europe · HORIZON-WIDERA-2024-TALENTS-03 · European Research Executive Agency (REA) · GA 101217448

- ▶ **Project Coordinator**; total budget ~€3 million. Widening/Talents action under Horizon Europe, focused on circular-economy capacity building.

POINGBIO — Knowledge Transfer Partnerships in biomedical polymer materials

2016 – 2022

Operational Programme Competitiveness 2014–2020 (ERDF) · Action 1.2.3 · P_40_443, SMIS 105689

- ▶ **Project Director**; total budget ~€3.2 million (RON 15.8M). Research-to-industry knowledge-transfer partnership.

As consortium / team member (selection)

- ▶ **FP6 Polysaccharides** — EPNOE Network of Excellence (NMP3-CT-2005-500375), 6th Framework Programme.
- ▶ **POSDRU** — “Cristofor C. Simionescu” postdoctoral fellowships, European Social Fund (**management team**).
- ▶ **National R&D programmes** — CNCS/UEFISCDI competitions (PCCA, PED, PN-II/III/IV), incl. experimental-demonstrative projects PERIOSILVERDDS and MH4DEVICE.
- ▶ In total, participation in 10+ national and international research projects across the main EU funding instruments.

AREAS OF EXPERTISE

- ▶ Polymer physics and chemistry; advanced and nanostructured polymer materials.
- ▶ Surface modification and functionalisation (plasma, chemical, UV and laser treatments).
- ▶ Biomaterials; biocompatible and antimicrobial surfaces (catheters, implants, wound dressings).
- ▶ Nanocomposites and nanoparticles (Ag, Au, iron oxides); electrospinning.
- ▶ Structural characterisation by X-ray diffraction (XRD/PXRD) and spectroscopy (RAMAN, FTIR).

- ▶ Materials for packaging, electromagnetic shielding, green synthesis and recycling (circular economy).
- ▶ Plasma physics and electrical discharges; plasma diagnostics.
- ▶ **Thematic alignment** with Horizon Europe priorities in advanced materials, health and circular economy.

EVALUATION & REVIEW COMPETENCIES

- ▶ **Peer reviewer** for international journals and publishers: **Elsevier, Wiley, MDPI, CRC Press – Taylor & Francis Group**.
- ▶ **Guest editor** of 5 special issues across 3 journals (ISI impact factor > 3); editor of 2 books.
- ▶ **Project Director and certified project manager** — direct command of proposal design, budgeting and monitoring, essential for assessing project feasibility and impact.
- ▶ **Member of Smart Specialisation (RIS3) evaluation bodies** — appraisal and monitoring of the North-East RIS3 strategy and of regional technology-transfer funding under EU Cohesion Policy.
- ▶ **Demonstrated applied/innovation expertise** through 4 patents and 2 patent applications and “Inventica” innovation awards — relevant to technology-transfer and impact assessment.

PROFESSIONAL EXPERIENCE

Senior Researcher, Grade I (CS I)

since 2026

“Petru Poni” Institute of Macromolecular Chemistry, Romanian Academy, Iași

- ▶ Leader of the group, project 9.2. “Physico-chemical properties of structured polymer materials” under Subprogramme 9 of the Romanian Academy programme “Macromolecular Compounds and Advanced Polymer Materials 2021–2027”.
- ▶ Role in scientific management, applied and fundamental research, experimental development and dissemination.

Senior Researcher, Grade II (CS II)

2017 – 2026

“Petru Poni” Institute of Macromolecular Chemistry, Romanian Academy, Iași

- ▶ Leader of the group, project 9.2 under Subprogramme 9; scientific management, research and experimental development, dissemination.

Researcher, Grade III (CS III) / Researcher

2006 – 2017

“Petru Poni” Institute of Macromolecular Chemistry, Romanian Academy, Iași

- ▶ Leader of the group, coordinator of the projects “Optimal conditions for polymer modification” (2008–2013) and “Nanostructuring of polymer surfaces by conventional and nonconventional methods” (2014–2020), Polymer Physics subprogramme.

PhD student, teaching assistant and associate lecturer

1998 – 2006

Faculty of Physics, “Al. I. Cuza” University, Iași

- ▶ Fundamental and applied research, teaching, coordination of student activities.

Physics teacher (tenured)

1996 – 1998

“Gh. Asachi” Arts and Crafts School Group, Iași

- ▶ Pre-university teaching.

EDUCATION & ACADEMIC TITLES

Habilitation certificate in Chemistry

2024

SCOSAAR, Romanian Academy

- ▶ Right to supervise doctoral candidates (doctoral mentoring).

PhD in Physics

2006

Faculty of Physics, “Al. I. Cuza” University, Iași

Engineering degree — Systems & Computer Science

2002

Faculty of Automatic Control and Computer Engineering, “Gh. Asachi” Technical University, Iași

Advanced studies diploma (Physics)

1997

Faculty of Physics, "Al. I. Cuza" University, Iași

BSc in Physics

1996

Faculty of Physics, "Al. I. Cuza" University, Iași

RELEVANT PROFESSIONAL TRAINING

- ▶ **Project Management and Digital Skills Development** — UNISMART, Padua, Italy (online), 2026.
- ▶ **Certified Project Manager and Trainer** — completion certificates, EUROFIN PROIECT, 2009.
- ▶ **Public Relations Specialist** — EUROFIN PROIECT, 2010.
- ▶ **Radiological safety, level 2** (GR field, CNCAN-authorized) — "Horia Hulubei" NIPNE, Bucharest, 2009.

RESEARCH STAYS, FELLOWSHIPS & SCHOOLS

- ▶ International research fellowships: **SOCRATES** — Aston University, Birmingham, UK (1999–2000) and **TEMPUS** — Université Paris-Sud, France (1996).
- ▶ **ICDD fellowships** — "Fundamentals" and "Advanced Methods in X-ray Powder Diffraction", International Centre for Diffraction Data, Pennsylvania, USA (2008).
- ▶ Specialisation schools: "Advances in Biomaterials" — Polish Academy, Vienna (2012); "Biomaterials: trends and perspectives" — "C. D. Nenițescu" Centre, Bucharest (2011); "Plasmas non thermiques et applications" (2003).

SCIENTIFIC OUTPUT

- ▶ **100+ articles** in Web of Science journals and **22 articles** in other recognised journals.
- ▶ **11 books and book chapters, 2 books as editor, 4 patents and 2 patent applications.**
- ▶ **h-index 24, ~1700 citations**; 44+ oral communications and 3 invited lectures.
- ▶ Supervision of 2 PhD candidates (since 2024).

ADVISORY & STRATEGIC ROLES (SMART SPECIALISATION)

Member of the Scientific Council

since 2020

"Petru Poni" Institute of Macromolecular Chemistry, Iași

Member of the Academic Advisory Committee

since 2017

Support to the North-East Regional Development Agency (RDA)

- ▶ Supporting the RDA in defining, revising and evaluating the North-East Smart Specialisation Strategy (RIS3).

Member of the Regional Innovation Consortium

since 2021

Advisory body coordinated by the North-East RDA

- ▶ Appraisal and monitoring of RIS3 North-East; endorsement of the regional framework for Priority Axis 1 — Technology Transfer, Regional Operational Programme.

LANGUAGES, AWARDS & MEMBERSHIPS

- ▶ **Languages:** English and French.
- ▶ **Awards:** multiple prizes at editions of the "Inventica" International Salon of Research, Innovation and Technology Transfer.
- ▶ **Professional memberships:** Romanian Chemical Society; International Centre for Diffraction Data (ICDD), USA.